

propellantless drag compensation by ambient-electron exchange

a measured thrust–power frontier and a scale-free feasibility condition — draft notes, aug 2026

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the gap: nN/mW corner is empty

cm- to dm-class spacecraft at 400–600 km lose altitude to drag on month-to-year timescales. keeping them alive needs continuous thrust of a few to tens of nN, from a mW-class power budget.

problem: no flight-proven EP system works here. the smallest controllable flight EP (FEEP/electrospray) sits 10^2 – $10^3\times$ above this demand on every axis — thrust, power, and mass.

- cold gas, PPT, electrospray, FEEP — all carry propellant in a tank, with feed lines, valves, or emitter arrays that add mass disproportionate to a cm-class body
- gridded-ion and hall thrusters are efficient only at 1–3 orders above this power budget
- the dry-system floor (tank + feed + valves + PPU $\approx$ 0.3–1 kg) doesn’t shrink to gram scale at any propellant load

proposal: fire electrons out through an aperture in a sealed conducting body. the body charges positive, drawing ambient ionospheric electrons back in. the ionosphere supplies both the neutralizing return current and the propellant — at ~ 60 ng/year. no tank, no valves, no neutralizer. every ion/hall thruster already carries the gun (the neutralizer).

operating principle

the cycle, with measured values from the 200 V reference run:

1. HV supply holds cathode at $-V$ relative to the body
2. electrons accelerate out through the aperture at energy $\approx e(V - \varphi)$ — measured mean exhaust 147.5 eV at $V = 200$ V; energy fraction $KE/(V - \varphi) = 0.81$ is constant across the voltage range
3. each departed electron leaves the body one charge positive; the float rises — measured: +17.0 V
4. the positive body attracts ambient ionospheric electrons collected on its conducting surfaces; in equilibrium collected current = escaped current (closes to 1.5–3.5% at all three operating points)
5. escaping beam carries net momentum $\rightarrow$ thrust. recollected electrons arrive at thermal-plus- $e\varphi$ from all directions, depositing small net momentum (gated per run, 1–9% of beam thrust)

no valves, no flow control, no gimbals, no plasma production stage. thrust is throttled by emission current and supply voltage alone.

simulation method & evidence discipline

simulation setup

parameter	value
code	WarpX v26.5 (GPU), RZ electrostatic
body	Ø10 mm × 5.9 mm closed can, 4 mm exhaust aperture, internal cathode disk
EB	conductor w/ charge-pump float update each step
ambient plasma	dayside 400 km: $n_e = 1.627 \times 10^{12} \text{ m}^{-3}$, $T_e = 1318.8 \text{ K}$, $kT_e = 113.6 \text{ meV}$
reservoir	infinite-ionosphere recycle (particle refill)
grid	$dr = dz = 0.15 \text{ mm}$ ($\lambda_D = 1.97 \text{ mm}$)
time step	CFL-derived
PPC	16 per species
ion mass	$400 m_e$ (reduced; caveat in limitations)
sim time	800 ns (≈ 29 capacitive settle times at reference)

evidence contract

- every run freezes + hashes its full physics config
- atomic manifest records config hash, git commit, random seed, code version, expected-vs-observed final iteration
- interrupted runs are discarded — never resumed
- acceptance gates (escape fraction, benign float, steady current balance, momentum bound, sheath containment, two ledger-vs-dump cross-checks at $\sim 10^{-9}$) fixed in versioned policy files *before* each run
- 100 V and 300 V scaling predictions, and competing collection-law hypotheses, pre-registered before those runs completed

validation ladder

below the production stages, run under the same contract:

1. analytic two-electrode laplace rung
2. thermal collection vs one-sided flux theory (closes to $\pm 1\%$)
3. biased collection at two potentials
4. bare floating body
5. electron-gun emission rungs

the measured thrust–power frontier

V [V]	I [mA]	φ [V]	F [nN]	P [mW]	F/P [$\mu\text{N}/\text{W}$]	escape	KE [eV]
100	0.121	+5.40	3.42	12.1	0.283	96.1%	77.2
200	0.342	+16.98	13.65	68.4	0.200	98.4%	147.5
300	0.630	+36.30	30.13	189	0.159	99.0%	210.1

the voltage frontier of the Ø10 mm chipsat at the dayside plasma row. all runs PASS on all pre-registered acceptance gates. each point holds the same emission ratio $I/I_{\text{CL}} = 1.46$. the 100 V and 300 V thrust values landed on their pre-registered predictions (3.4 and 30 nN) to better than 1%. grid-resolution uncertainty:

$F \pm 4\%$, $KE \pm 7\%$, $\varphi \pm 2\%$; *particle count, time step, and seed contribute $\leq 0.5\%$.*

two efficiencies (kept side by side on purpose)

	energy conversion	impulse economy
value	$\eta = P_{\text{jet}}/P_{\text{supply}} \approx 0.73$	$F/P \approx 0.2 \mu\text{N}/\text{W}$ at 200 V
character	ion-thruster-class, nearly flat in V	$\sim 200\times$ below gridded ion, $\propto 1/\sqrt{V}$
meaning	superb energy-to-jet conversion	tiny momentum per watt

both are true simultaneously *because the exhaust is electrons*. the second number is why the device only owns the nN regime; the first is why it deserves to.

throttle law

the frontier confirms $P \propto F\sqrt{V}$ at fixed thrust: every doubling of voltage costs $\sim 40\%$ more power. so **slow-and-many beats fast-and-few** on momentum per watt, and the cheapest operating point is the lowest voltage whose emission ceiling still meets the thrust demand.

collection law: pre-registered discrimination

every escaping electron must be replaced by a collected ambient one. the float φ is whatever makes that balance close.

collected current:

$$I = \beta A j_{\text{the}}(n_e, T_e) (1 + \chi)^\alpha, \quad \chi = \frac{e\varphi}{kT_e}$$

three candidate exponents existed in the project lineage. all agree at the 200 V anchor ($\chi \approx 150$), diverge at 300 V ($\chi \approx 320$). predictions for the 300 V float recorded in the repo *before* the run:

law	α	predicted $\varphi(300 \text{ V})$	verdict
linear (orbital-motion style)	1.0	+31 V	refuted
sub-linear (fitted, predecessor)	0.82 ± 0.06	+36 V	survives
square root	0.5	+90 V	refuted (would fail benign gate)
measured	—	+36.30 V	

- the linear law is refuted — float passes 31 V and keeps climbing
- the square-root law is refuted — nothing in the trajectory approaches 90 V
- $\alpha = 0.82$ survives
- least-squares fit across all three points: $\alpha = 0.85\text{--}0.89$

caveat (pre-registered with the test): at high χ the sheath equilibrates on the ion timescale. the 300 V run's late slope, though decaying ($+44 \rightarrow +27 \text{ mV/ns}$ over the final 200 ns), had not reached zero. settled-value extrapolation: $\approx 42\text{--}48 \text{ V}$, brushing the 50 V design limit.

the float robs the beam — the accelerating potential is always $V - \varphi$ — so this law is the **charging tax schedule**: φ eats 5% of the supply at 100 V and 12% at 300 V.

throttle principle & ideal power bound

in the ideal limit (every emitted electron escapes at full energy, $\varphi \ll V$), the device economics contain **no free parameters at all**:

$$F = \sqrt{\frac{2m_e}{e}} I \sqrt{KE}, \quad c_{\text{ideal}} = \sqrt{\frac{2m_e}{e}} = 3.37 \text{ nN}/(\text{mA} \cdot \sqrt{\text{eV}})$$

fixed-thrust power law: $P = \frac{F \sqrt{V}}{c}$

this is a geometry-free theoretical lower bound. the minimum-power throttle uses the largest feasible escaped current and the lowest acceleration voltage that satisfies the thrust command.

how close does the measured geometry sit?

constant	value	meaning
c_F	$3.27 = 0.97 c_{\text{ideal}}$	thrust slope (3% deficit is plume divergence)
κ	0.81	exhaust-energy fraction (injection-plane space-charge depression)
c_{eff}	$= c_F \sqrt{\kappa} = 0.87 c_{\text{ideal}}$	effective constant

with this calibration the law reproduces the gated frontier powers to within 4–6%:

V [V]	model P [mW]	measured P [mW]	ratio
100	11.7	12.1	1.03
200	65.8	68.4	1.04
300	177.9	189	1.06

the residual is the charging tax $V/(V - \varphi) = 1.06$ – 1.14 .

a concrete geometry already operates within $\approx 20\%$ of the parameter-free physics bound (measured/bound = 1.19–1.22 across the frontier), for understood, separable reasons.

momentum ledger: is the thrust real?

the first question anyone asks: the emitted current *returns* — does its momentum cancel the thrust?

predecessor implementation ($\varnothing 14$ mm, 800 V)

closed the full control-volume momentum balance every sim window: per-species momentum flux through every boundary, deposition on the body, emission reaction, field terms ($\int \rho E_z dV$, maxwell stress), and the stored-momentum transient.

two validation fixtures passed pre-registered gates:

fixture	what it tests	result	gate
vacuum escape	analytic answer	residual +0.80 nN	$< 1 \text{ nN}$ ✓
full-return null	body pinned above beam energy, escape = 0%	impact: -12.1 nN ; closed: $-0.39 \text{ nN} \approx 0$	✓

at the operating point: ledger closes at $F = F_{\text{beam}} - 2.3\%$. the return-current impact channel is cancelled by the field pull on the body — the naive subtraction $F_{\text{beam}} - F_{\text{impact}}$ *underestimates* thrust.

present campaign (gated per run)

V [V]	$ F_{\text{net}} /F_{\text{beam}}$	note
100	8.8%	slower beam, lower escape $\rightarrow$ return channel relatively largest
300	1.0%	0.30 nN vs 30.13 nN

physics ceiling: collected electrons arrive with $\sim e\varphi$ of directed energy against the beam's $e \cdot \text{KE}$. even if every collected electron arrived from directly astern, the cancellation could not exceed $\sqrt{\varphi/\text{KE}} \approx 41\%$ (300 V values). isotropic arrival reduces it to the measured percent level.

numerical credibility

axis	test	result
code-to-code	migrated GPU vs predecessor, 200 V anchor	φ to 0.05%, escape to 0.001%, F to 0.09%
collection physics	thermal rung vs one-sided flux theory	$\pm 1\%$
ledger cross-check	per-step scrape vs independent particle dumps	agreement at 10^{-9} (gate: 2×10^{-2})
determinism	seeded reruns	exact reproduction
PPC $\times 2$	ambient + beam doubled (this work)	all metrics $\leq 0.05\%$ — closed axis
dt , seed	$\times 2$ finer, different seed (predecessor)	$\leq 0.5\%$ each
grid 0.15 $\rightarrow$ 0.10 mm	matched 430–536 ns window	$F + 4.0\%$, $\text{KE} + 7.4\%$, $\varphi - 1.8\%$

grid resolution is the leading numerical uncertainty. carried as the error band on all frontier numbers ($F \pm 4\%$, $\text{KE} \pm 7\%$, $\varphi \pm 2\%$). the shift's sign is conservative for every gate: finer grids give more thrust and a lower float. a third grid point to establish the asymptotic order is future work.

comparison with existing micropropulsion

related work

the idea of exchanging charge with the ambient plasma has deep roots, though not in exactly this form:

- **bare-tether OML collection** (sanmartín et al. 1993) — theoretical foundation for our collection physics. flown + confirmed by NASA's plasma motor generator (PMG) tether experiment, 1993
- **electric solar wind sail** (janhunen 2004) — continuous electron gun holds tether positive, deflects solar-wind protons. different mechanism (electrostatic deflection vs direct beam recoil) but shares the defining feature: electron emission makes it propellantless
- **plasma brake** (janhunen 2010) — negative-polarity tether for coulomb drag deorbiting
- **dipole drive** (zubrin 2017) — accelerates collected ions between charged screens, targets magnetospheric regimes

- **ABEP / atmosphere ion thrusters** — collect neutral atmosphere or ram ions and ionize/accelerate with a conventional EP stage. our device requires none of these stages

at km-scale, per watt, electrodynamic-tether thrust beats any reaction engine including ours. the comparison inverts at cm-scale: a 5 mm vehicle vs a deployed conductor 10^4 – $10^6\times$ its own length, and a bare tether still needs electron collection and emission at its ends — the same contactor physics this device *is*, minus the tether.

quantitative comparison

system	thrust	power	I_{sp}	η_{jet}	hardware
this work	3.4–30 nN	12–189 mW	0.7 – 1.5×10^6 s	68–73%	HV supply + gun + aperture
cold gas	0.1–10 mN	<2 W	60–110 s	—	dual tanks, valves, heaters, nozzles
PPT	40 μ N avg	$\sim$ 2 W	655 s	<20%	chamber, cap bank, EMI box
electrospray	4×20 μ N	16.5 W	225 s	28/45%	CNT neutralizer, tanks, PPU
FEED	10 μ N–0.5 mN	8–40 W	2k–5k s	$\sim$ 35%	28-needle emitter, 2 neutralizers, PPU
RF ion	0.66–1.24 mN	56–80 W	1.4k–2.6k s	—	grid optics, iodine feed, PPU, gimbal
hall	13 mN	200–250 W	1390 s	$\sim$ 44%	discharge channel, hollow cathode, PPU
solar pressure	$\sim$ 91 nN	passive	—	—	none; no throttling, no eclipse

sources: NASA state of the art of small spacecraft technology 2026, manufacturer datasheets.

three takeaways:

1. every incumbent’s lowest flight power is ≥ 1 order of magnitude above our 300 V ceiling (189 mW)
2. our energy conversion (73%) exceeds every value found (28–45% range)
3. our F/P is $\sim 200\times$ below gridded ion — the efficiency-indifference boundary is real. the F/P penalty is invisible at nN (mW), noticeable at μ N (~ 10 W), decisive at mN

the natural handoff — this device below ~ 0.1 μ N, electrospray at μ N, ion/hall at mN — is part of the claim, not a concession.

mission feasibility corridor

the three gated runs calibrate a minimal executable model: per 5-minute timestep, local plasma density + temperature (IRI-2020) and drag (NRLMSISE-00, real F10.7/Ap, TudatPy propagation) enter, and the model solves for the minimum-power operating point (V, I, φ) self-consistently with the measured collection law. any row outside what a committed run has measured is flagged extrapolated, never silently averaged in.

year-long mission sweeps

altitude / pose	drag mean/max [nN]	feasible	duty cycle	P mean [mW]	$\varphi > 50$ V	in envelope
400 km axial	32.9 / 92.4	21%	140%	136	41%	12%
400 km lateral	21.7 / 60.7	54%	92%	111	31%	33%
500 km axial	7.6 / 28.4	81%	45%	39	19%	29%
550 km axial	3.8 / 16.3	92%	32%	17	8%	8%
600 km axial	2.0 / 9.6	97%	25%	8	3%	1%

“feasible” = a benign equilibrium exists at demanded thrust. “duty cycle” = mean drag / mean benign capability (>100% cannot close). “in envelope” = row interpolates committed measurements on every axis.

the corridor reads directly:

- **550–600 km:** drag compensation closes on body-mounted solar power (8–17 mW mean demand vs ~30 mW harvest), 25–32% duty cycles
- **500 km:** closes on impulse but power-bound at the mean (39 mW) — closure relies on thrusting preferentially when plasma is dense and charging tax low
- **400 km: does not close** for this body — demand exceeds emission ceiling 63% of the time

honesty flags:

1. the corridor’s boundary behaviour is set by night-side rows (thin, cold plasma → float toward 50 V limit) — exactly the rows the PIC has not measured. a single night-row run is the targeted next measurement
2. power here is beam power IV ; converter and cathode overheads are not modelled

geometry lever: slender bodies

drag and collection scale with *different* areas: drag buys the ram silhouette, collection and solar harvest buy the total skin. a slender body flying end-on decouples them.

example: $\varnothing 10\text{ mm} \times 100\text{ mm}$ cylinder — same end cap as chipsat (same nN-class drag), lateral skin $\sim 31\text{ cm}^2$ ($10\times$ chipsat surface), collects $\sim 46\text{ }\mu\text{A}$ of bare thermal current, carries enough solar cell area for 100–200 mW, mass tens of grams.

pre-registered shape test

concern: the collection exponent is geometry-dependent (OML: $\alpha = 1$ for sphere, $\alpha = 0.5$ for long cylinder). our squat can measured 0.82–0.89. a slender body should slide toward the cylinder value — weaker enhancement per volt, hidden charging tax.

two competing hypotheses pre-registered and measured ($\varnothing 10\text{ mm} \times 30.5\text{ mm}$ can, $L/r = 6$, skin $3.24\times$ chipsat’s, at identical 200 V drive):

hypothesis	prediction	measured	verdict
A: area-only (α holds)	$\varphi \approx 4\text{--}5\text{ V}$	$2^*\varphi = 4.38\text{ V}$ (tail-avg)	confirmed
B: cylinder-limit wall	φ tens of V		refuted by $\sim 10\times$

all six required gates pass. escape holds at 98.4%.

the fitted exponent survives a $3.24\times$ area change and an aspect-ratio change from $L/r = 0.6$ to 6 .

elongation is free on thrust side: because $\text{KE} = \kappa(V - \varphi)$, floating $3.9\times$ lower keeps more drive — exhaust energy rises from ~ 147 to 159.7 eV , beam thrust from 13.6 to 14.22 nN , at the same current and same ram silhouette.

design trades:

- solar cells and bare collector compete for the same wall area (clad dielectric surfaces don't collect)
- end-on flight isn't attitude-free (gravity gradient prefers long axis nadir-pointed = side-on to ram)
- emission ceiling scales as $1/d^2$ — a slender body must keep the cathode–aperture gap at its short design value and grow the body *around* the gun

size cancels: scale-free feasibility

the geometry measurement licenses a scaling argument. drag charges for A_{ram} , so thrust demand, beam current, and input power are all $\propto A_{\text{ram}}$. solar harvest $\propto A_{\text{skin}}$. **vehicle size cancels from both sides of the power balance.** what survives: the shape ratio $A_{\text{skin}}/A_{\text{ram}}$ + altitude.

		400 km	500 km	550 km	600 km
power demand per m ² of ram		1451 W	335 W	170 W	88 W
<i>margin = harvest / demand at 100 V; >1 closes</i>					
Ø10 mm can, squat	(4)	0.1	0.4	0.8	1.6
1U cube, face-on	(6)	0.1	0.6	1.2	2.3
Ø10 mm can, slender	(14)	0.3	1.4	2.8	5.4
3U CubeSat, end-on	(14)	0.3	1.4	2.8	5.4
6U CubeSat, end-on	(12)	0.3	1.2	2.4	4.6

harvest assumes 30% cells over $\frac{1}{3}$ skin at 25% orbit-average illumination (102 W/m² of cell area).

the slender Ø10 mm can and the 3U CubeSat return **identical margins** because they have the same shape ratio. the corridor of the mission table is not a chipsat result — it's a statement about aspect ratio + altitude that a payload-carrying spacecraft inherits unchanged.

absolute values for a 3U end-on:

- 600 km: 8.8 mA at 0.88 W
- 550 km: 17 mA at 1.7 W
- 500 km: 33 mA at 3.4 W

two things get *easier* with size:

1. larger skins collect more → enhancement demanded over bare thermal collection falls (3U needs $4.3\times$ thermal flux at 600 km where chipsat anchor needs $15\times$). **larger bodies rest on less extrapolation, not more**
2. lower $\chi \rightarrow$ lower float $\rightarrow$ more drive energy stays in the beam

unmeasured regime: all committed runs sit at $r/\lambda_D \approx 2.5$. CubeSat radii are 25–60 λ_D — a thin-sheath regime where OML doesn't apply. child-sheath estimate: 3U floats ~ 12 V at 600 km, ~ 25 V at 550 km, ~ 47 V at 500 km — benign in the corridor, tightening at its lower edge. this is an estimate, not a calibration. a single large-body run would settle it.

practical realization

bill of materials

- a conducting structure
- a milliwatt-class HV supply (100–300 V; up to 800 V demonstrated at larger scale)
- a cold electron gun with cathode at $-V$
- an aperture

no tank, valves, feed lines, RF stages, or gimbals. the dry-system floor is a cathode, a boost converter, and the spacecraft’s own skin.

retrofit path

gridded-ion and hall thrusters already fly electron sources as neutralizers. after propellant exhaustion, the neutralizer + the spacecraft structure *is* this thruster: a software-and-wiring change buys drag-makeup-class thrust for mission life extension.

limitations & future work

limitation	status	impact
collection law density axis unmeasured	all 3 frontier runs share 1 day-side row	night-row run is highest-value next measurement
collection law geometry axis: 2 points	exponent survives to $L/r = 6$ and $3.24\times$ skin	beyond that remains extrapolation
no CubeSat sheath regime	all bodies at $r/\lambda_D \approx 2.5$; CubeSats are 25–60	child-sheath estimate, not calibration; single large-body run would settle it
grid resolution	$0.15 \rightarrow 0.10$ mm: $F + 4\%$, KE $+7\%$	leading uncertainty; 3rd grid point needed
finite-time equilibrium	300 V float still relaxing at run end	settled band 42–48 V carried through α fit
reduced ion mass	$400 m_e$ vs real O^+ ($29\,166 m_e$)	float sensitivity to ion mass not measured
stationary plasma	no bulk flow modelled	ram-ion wake is largest unmodelled effect
2D axisymmetry, no B field	RZ; real flight sees transverse B	deferred
system power	beam power IV only	cathode heating + converter efficiency are flight-design quantities
throttle optimization deferred	off-optimum: $1.5\text{--}2\times$ bound	closed-loop controller design is future engineering
full mission sim	no battery SOC, seasonal density, attitude coupling	per-row accounting is the input such a study needs

conclusion

drag compensation for small spacecraft at 450–600 km requires nN to μ N of continuous thrust from mW to a few W — a slot no flight propulsion system occupies.

three results support treating this as a credible propulsion mechanism:

1. **the frontier is measured:** 3.4 nN at 12 mW to 30 nN at 189 mW across the full hardware voltage range. $F/P \propto 1/\sqrt{V}$ confirmed within 2.5%. energy conversion ≈ 0.73 at $F/P \sim 200\times$ below gridded ion — the pair of numbers that assigns this device the nN regime and nothing above it
2. **the thrust is real:** closed momentum ledger with full-return null experiment (predecessor); per-run momentum gates bound return channel at 1–9% of beam thrust (this campaign)
3. **the mission corridor is honest:** closure on harvested power at 550–600 km, impulse closure at 500 km, failure at 400 km stated plainly. size cancels: a 3U CubeSat and a $\varnothing 10$ mm slender can share identical margins at the same shape ratio

the device is cheap to test. no hardware beyond an HV supply and an electron gun of a kind that already flies as a neutralizer. the concept also offers a retrofit path: after propellant exhaustion, run the neutralizer alone for drag-makeup-class thrust.

this is a feasibility-concept study: the device physics is measured and gated, the mission projection is model-based with every extrapolated regime explicitly flagged, and flight validation is future work.